NILEBRIDGE PROFESSIONAL ACADEMY

Data Analytics & Business Intelligence

DEMONSTRATION CATALOGUE · 2026–2027 · CATALOGUE CODE DA

CATALOGUE OVERVIEW

A practical catalogue for spreadsheet analysis, SQL querying, and Python-based analytics. The programmes emphasize data validation, metric definitions, reproducible transformations, and clear business communication.

ACADEMY AND DATA NOTICE

NILEBRIDGE PROFESSIONAL ACADEMY is a fictional demonstration institution. All programme records, durations, curriculum details, assessments, and career descriptions are synthetic examples created for RAG ingestion and retrieval testing. They are not real academy offerings, verified schedules, official accreditation claims, or employment guarantees.

Programme Directory

The programme records below use stable programme identifiers to support retrieval and metadata validation.

Programme ID	Programme title	Level	Duration
DA-XL-101	Spreadsheet Analytics & Reporting	Foundation	45 guided learning hours
DA-SQL-201	SQL for Analytics & Reporting	Foundation to Intermediate	60 guided learning hours

Information included for each programme

- Programme identity, level, duration, delivery format, and intended audience
- Entry requirements and programme purpose
- Learning outcomes and week-by-week curriculum
- Practical labs, assessment weights, and grading criteria
- Capstone brief, deliverables, tools, and FAQ
- Provenance and accreditation status

PROGRAMME 01 · DA-XL-101

Spreadsheet Analytics & Reporting

Indicative hours	Duration	Level	Delivery
45 guided learning hours	6 weeks	Foundation	Workshops and spreadsheet labs.

Programme summary

Develops a reliable workflow for cleaning tabular data, using formulas, building pivot summaries, and presenting results through clear dashboards.

Intended audience

Beginners and business users who work with structured data.

Entry requirements

Basic computer literacy.

Learning outcomes

- Prepare consistently structured tables and apply validation rules.
- Use formulas, lookups, conditional aggregation, and pivot tables.
- Identify duplicates, missing values, and inconsistent categories.
- Build charts with labels, units, and documented metric definitions.

Week-by-week curriculum

Week	Topic	Coverage	Practice / evidence
1	Tables and data types	Structure records, headers, dates, numbers, and categories.	Create a data dictionary.
2	Formulas and lookups	Apply logical, text, date, lookup, and conditional formulas.	Complete a formula practice workbook.
3	Data quality	Inspect missing values, duplicates, invalid dates, and inconsistent labels.	Create a data-quality log.
4	Pivot analysis	Group and summarize measures by category and period.	Build a pivot-based summary.
5	Charts and dashboards	Select charts, label metrics, and arrange a clear dashboard.	Review and revise a dashboard.
6	Capstone	Combine cleaning, analysis, and reporting.	Submit a workbook and one-page findings note.

Practical labs

- Formula practice
- Duplicate and missing-value review
- Pivot-table analysis
- Dashboard critique

Assessment structure

Component	Weight	Evidence / criteria
Exercises	30%	Formula correctness and data validation.
Dashboard	30%	Clear layout, appropriate chart selection, and metric definitions.
Capstone	40%	Reproducibility, correctness, and findings communication.

Capstone project

Prepare a sample sales or operations workbook with data-quality checks, grouped metrics, a dashboard, and a short report explaining three findings and their limitations.

Suggested grading rubric

- Data validation: 25%

- Analysis correctness: 30%
- Dashboard clarity: 25%
- Documentation: 20%

Tools and technologies

Microsoft Excel or compatible spreadsheet application · CSV

Frequently asked questions

Is advanced statistics required?

No. The programme focuses on practical spreadsheet analysis.

Should duplicates always be deleted?

No. First determine whether they are true duplicate records or legitimate repeated transactions.

What makes a KPI useful?

A clear definition, calculation, scope, time window, and connection to a decision.

Record and provenance

Department: Data Analytics | Skill level: Foundation | Accreditation: not verified | Country: not specified | Source status: synthetic demonstration data | Catalogue: Data Analytics & Business Intelligence | Programme: Spreadsheet Analytics & Reporting | Source file: 02_Data_Analytics_BI_Expanded_Catalogue.pdf.

PROGRAMME 02 · DA-SQL-201

SQL for Analytics & Reporting

Indicative hours	Duration	Level	Delivery
60 guided learning hours	8 weeks	Foundation to Intermediate	Instructor-led query labs and case-based exercises.

Programme summary

Builds the ability to retrieve, join, aggregate, and validate relational data to answer practical reporting questions.

Intended audience

Aspiring analysts and business users who need to query relational data.

Entry requirements

Basic computer literacy; no previous SQL required.

Learning outcomes

- Write SELECT queries with filters, sorting, and conditional logic.
- Aggregate records and use joins without accidentally multiplying counts.
- Use subqueries, CTEs, and window functions for analytical tasks.
- Validate query results and document metric logic.

Week-by-week curriculum

Week	Topic	Coverage	Practice / evidence
1	Relational foundations	Tables, keys, relationships, and schema reading.	Diagram a small database.
2	Filtering and sorting	SELECT, WHERE, ORDER BY, CASE, and null behavior.	Answer basic data questions.
3	Aggregation	COUNT, SUM, AVG, GROUP BY, and HAVING.	Create a grouped KPI report.
4	Joins	INNER/LEFT joins, keys, and row-multiplication checks.	Join three related tables.
5	Subqueries and CTEs	Break complex queries into readable stages.	Refactor a long query.
6	Window functions	Ranking, running totals, and partitioned calculations.	Create a ranking report.
7	Quality and performance	Control totals, query readability, and basic indexing concepts.	Audit a flawed query.
8	Capstone	Build and validate an analytical query pack.	Present results and definitions.

Practical labs

- Filtering and aggregation drills
- Join cardinality checks
- Window-function analysis
- Metric validation exercise

Assessment structure

Component	Weight	Evidence / criteria
Query labs	30%	Correct SQL and readable structure.
Practical test	20%	Join logic, null handling, and aggregation.
Capstone	50%	Useful questions, validated outputs, and documented definitions.

Capstone project

Use a sample relational schema to answer ten business questions, include commented SQL, verify row counts and totals, and summarize the business interpretation.

Suggested grading rubric

- SQL correctness: 35%
- Validation and join logic: 25%
- Metric definitions: 20%
- Communication: 20%

Tools and technologies

SQL · SQLite or PostgreSQL · database client

Frequently asked questions

Why can a join inflate totals?

A join can produce multiple output rows for one source record when the join key is not unique on one or both sides.

When should I use HAVING instead of WHERE?

WHERE filters rows before grouping; HAVING filters grouped results after aggregation.

Are all SQL dialects identical?

No. Core concepts transfer, but functions and syntax vary between database platforms.

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